ROBOTICS 4/M

TOPIC 8: DYNAMICS AND FORCES

8.1 INTRODUCTION

In order for robot manipulators to produce enough force to perform the task they are required to do, the joints must produce enough combine force to accomplish the desired task. The dynamics of the robot and its joints are defined in terms of the resulting accelerations that are caused by the joint forces. Both the dynamics and the accelerations are related to the loads, masses and inertias of the joints and links [1,2].

In order for a mass m (e.g. a robot link) to be accelerated (by a) a force F must be exerted on the mass. This also applies to the rotational dynamics of a rotating body i.e. and angular acceleration α is applied by a torque T [1]:

$$\sum F = m.a \text{ and } \sum T = I.\alpha \tag{8.1}$$

In order to accelerate a robot's link it is required to have the joint actuators exert sufficient force and/or torque on the link so that it moves with the desired acceleration and velocity. If the joint does not move the link with sufficient force/torque, then the link will not move quickly enough and the robot will lose positional accuracy.

The analysis of the dynamic forces/torques acting on a body can be performed using Newtonian mechanics where the sum of all the forces acting on a link are summed to form a dynamic equation for the robot. This becomes very complex when you consider that the robot is a multiple degree of freedom mechanism with distributed masses. Instead the dynamics analysis of robot manipulators id performed using *Lagrangian Mechanics*.

8.2 LAGRANGIAN MECHANICS

Lagrangian mechanics is based on the differentiation of energy relationships with respect to the system's variables and time. The initial stage of the analysis is to form something called the *Lagrangian*, L, which is the difference between kinetic energy E_K and the potential energy E_P of the system i.e. [1,2].

$$L = E_K - E_P = K - P \tag{8.2}$$

The summation of the external forces acting on the system as a result of a linear motion is represented thus:

$$F_i = \frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{x}_i}\right) - \frac{\partial L}{\partial x_i} \tag{8.3}$$

The summation of the external forces acting on the system as a result of a rotational motion is represented thus:

$$T_i = \frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{\theta}_i}\right) - \frac{\partial L}{\partial \theta_i} \tag{8.4}$$

Therefore, in order to derive the equations of motion of a robot (or any other system) we must define the kinetic and potential energy relationships for the system. Some examples are given below:

MASS-SPRING SYSTEM

A mass-spring system shown in Figure 8.1 [1].

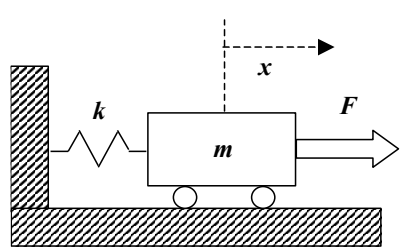

Figure 8.1: Mass-Spring System

This system can be defined in terms of its kinetic and potential energy:

$$K = \tfrac{1}{2} m v^2 = \tfrac{1}{2} m \dot{x}^2 \tag{8.5}$$

$$P = \tfrac{1}{2} k x^2 \tag{8.6}$$

The differentiation of the Lagrangian yields the following linear force equation:

$$F = m\ddot{x} + kx \tag{8.7}$$

MASS-SPRING-PENDULUM

A mass-spring-pendulum system shown in Figure 8.2 [1].

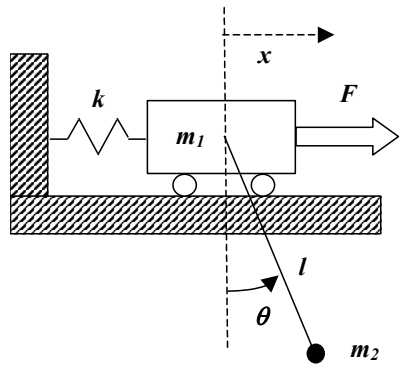

Figure 8.2: Mass-Spring-Pendulum System

This system can be defined in terms of its kinetic and potential energy:

$$K = \tfrac{1}{2}(m_1 + m_2)\dot{x}^2 + \tfrac{1}{2} m_2 \left(l^2 \dot{\theta}^2 + 2l\dot{\theta}\dot{x}\cos\theta\right) \tag{8.8}$$

$$P = \tfrac{1}{2} k x^2 + m_2 g l (1 - \cos\theta) \tag{8.9}$$

The differentiation of the Lagrangian yields the following linear force equation:

$$F = (m_1 + m_2)\ddot{x} + m_2 l \ddot{\theta}\cos\theta - m_2 l \dot{\theta}^2 \sin\theta + kx \tag{8.10}$$

and the following rotational torque equation:

$$T = m_2 l^2 \ddot{\theta} + m_2 l \ddot{x}\cos\theta + m_2 g l \sin\theta \tag{8.11}$$

These equations can be written in the following matrix form:

$$\begin{bmatrix} F \\ T \end{bmatrix} = \begin{bmatrix} m_1 + m_2 & m_2 l\cos\theta \\ m_2 l\cos\theta & m_2 l^2 \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} 0 & m_2 l\sin\theta \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{x}^2 \\ \dot{\theta}^2 \end{bmatrix} + \begin{bmatrix} kx \\ m_2 g l \sin\theta \end{bmatrix} \tag{8.12}$$

TWO-LINK MECHANISM

A two-link mechanism system is shown in Figure 8.3 [1].

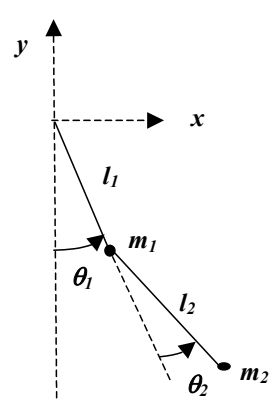

Figure 8.3: Two-Link Mechanism

This system can be defined in terms of its kinetic and potential energy:

$$K = \tfrac{1}{2}(m_1 + m_2)l_1^2\dot{\theta} \tag{8.13}$$

$$P = -(m_1 + m_2)gl_1\cos\theta_1 - m_2 g l_2 \cos(\theta_1 + \theta_2) \tag{8.14}$$

The differentiation of the Lagrangian yields the following rotational torque equations for m_1 and m_2:

$$\begin{aligned} T_1 = &\left[(m_1 + m_2)l_1^2 + m_2 l_2^2 + 2m_2 l_1 l_2 \cos\theta_2\right]\ddot{\theta} \\ &-m_2 l_1 l_2 \sin\theta_2 \dot{\theta} \qquad m_2)gl_1 \sin\theta_1 + m_2 g l_2 \sin(\theta_1 + \theta_2) \end{aligned} \tag{8.15}$$

$$T_2 = \left[m_2 l_2^2 + m_2 l_1 l_2 \cos\theta_2\right]\ddot{\theta}_1 + m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \sin\theta_2 \dot{\theta}_1^2 + m_2 g l_2 \sin(\theta_1 + \theta_2) \tag{8.16}$$

These equations can be written in the following matrix form:

$$\begin{aligned} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = &\begin{bmatrix} (m_1 + m_2)l_1^2 + m_2 l_2^2 + 2m_2 l_1 l_2 \cos\theta_2 & m_2 l_2^2 + m_2 l_1 l_2 \cos\theta_2 \\ m_2 l_2^2 + m_2 l_1 l_2 \cos\theta_2 & m_2 l_2^2 \end{bmatrix}\begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} \\ &+\begin{bmatrix} 0 & -m_2 l_1 l_2 \sin\theta_2 \\ m_2 l_1 l_2 \sin\theta_2 & 0 \end{bmatrix}\begin{bmatrix} \dot{\theta}_1^2 \\ \dot{\theta}_2^2 \end{bmatrix} + \begin{bmatrix} -m_2 l_1 l_2 \sin\theta_2 & -m_2 l_1 l_2 \sin\theta_2 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} \dot{\theta}_1\dot{\theta}_2 \\ \dot{\theta}_2\dot{\theta}_1 \end{bmatrix} \\ &+\begin{bmatrix} (m_1 + m_2)gl_1 \sin\theta_1 + m_2 g l_2 \sin(\theta_1 + \theta_2) \\ m_2 g l_2 \sin(\theta_1 + \theta_2) \end{bmatrix} \end{aligned} \tag{8.17}$$

TWO-DEGREE-OF-FREEDOM ROBOT ARM

A two-degree-of-freedom robot arm is shown in Figure 8.4 [1].

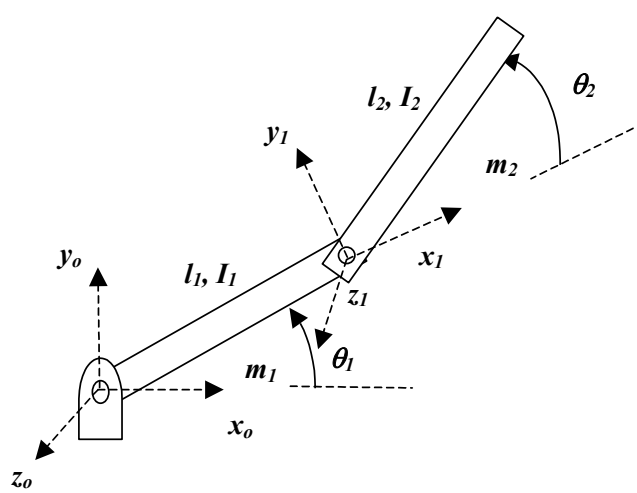

Figure 8.4: Two-degree-of-freedom robot arm

This system can be defined in terms of its kinetic and potential energy:

$$\begin{aligned} K = &\dot{\theta}_1^2\left(\frac{1}{6}m_1 l_1^2 + \frac{1}{6}m_2 l_2^2 + \frac{1}{2}m_2 l_1^2 + \frac{1}{2}m_2 l_1 l_2 C_2\right) \\ &+\dot{\theta}_2^2\left(\frac{1}{6}m_2 l_2^2\right) + \dot{\theta}_1\dot{\theta}_2\left(\frac{1}{3}m_2 l_2^2 + \frac{1}{2}m_2 l_1 l_2 C_2\right) \end{aligned} \tag{8.18}$$

$$P = m_1 g \frac{l_1}{2} S_1 + m_2 g\left(l_1 S_1 + \frac{l_2}{2} S_{12} \right) \tag{8.19}$$

The differentiation of the Lagrangian yields the following rotational torque equations for m_1 and m_2:

$$\begin{aligned} T_1 = & \left(\frac{1}{3} m_1 l_1^2 + m_2 l_1^2 + \frac{1}{3} m_2 l_2^2 + m_2 l_1 l_2 C_2 \right) \ddot{\theta}_1 + \left(\frac{1}{3} m_2 l_2^2 + \frac{1}{2} m_2 l_1 l_2 C_2 \right) \ddot{\theta}_2 \\ & - \left(m_2 l_1 l_2 S_2 \right) \dot{\theta}_1 \dot{\theta}_2 - \left(\frac{1}{2} m_2 l_1 l_2 S_2 \right) \dot{\theta}_2{}^2 \\ & + \left(\frac{1}{2} m_1 + m_2 \right) g l_1 C_1 + \frac{1}{2} m_2 g l_2 C_{12} \end{aligned} \tag{8.20}$$

$$T_2 = \left(\frac{1}{3} m_2 l_2^2 + \frac{1}{2} m_2 l_1 l_2 C_2 \right) \ddot{\theta}_1 + \left(\frac{1}{3} m_2 l_2^2 \right) \ddot{\theta}_2 + \left(\frac{1}{2} m_2 l_1 l_2 S_2 \right) \dot{\theta}_1{}^2 + \frac{1}{2} m_2 g l_2 C_{12} \tag{8.21}$$

These equations can be written in the following matrix form:

$$\begin{aligned} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = & \begin{bmatrix} \frac{1}{3} m_1 l_1^2 + m_2 l_1^2 + \frac{1}{3} m_2 l_2^2 + m_2 l_1 l_2 C_2 & \frac{1}{3} m_2 l_2^2 + \frac{1}{2} m_2 l_1 l_2 C_2 \\ \frac{1}{3} m_2 l_2^2 + \frac{1}{2} m_2 l_1 l_2 C_2 & \frac{1}{3} m_2 l_2^2 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} \\ & + \begin{bmatrix} 0 & -\frac{1}{2} m_2 l_1 l_2 S_2 \\ \frac{1}{2} m_2 l_1 l_2 S_2 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1{}^2 \\ \dot{\theta}_2{}^2 \end{bmatrix} + \begin{bmatrix} -\frac{1}{2} m_2 l_1 l_2 S_2 & -\frac{1}{2} m_2 l_1 l_2 S_2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \dot{\theta}_2 \\ \dot{\theta}_2 \dot{\theta}_1 \end{bmatrix} \\ & + \begin{bmatrix} \left(\frac{1}{2} m_1 + m_2 \right) g l_1 C_1 + \frac{1}{2} m_2 g l_2 C_{12} \\ \frac{1}{2} m_2 g l_2 C_{12} \end{bmatrix} \end{aligned} \tag{8.22}$$

8.3 THE EQUATIONS OF MOTION

The equations of motion for the above examples (i.e. Equations (8.12), (8.17) and (8.22)) can be written in the following symbolic form [1]:

$$\begin{bmatrix} T_i \\ T_j \end{bmatrix} = \begin{bmatrix} D_{ii} & D_{ij} \\ D_{ji} & D_{jj} \end{bmatrix} \begin{bmatrix} \ddot{\theta}_i \\ \ddot{\theta}_j \end{bmatrix} + \begin{bmatrix} D_{iii} & D_{ijj} \\ D_{jii} & D_{jjj} \end{bmatrix} \begin{bmatrix} \dot{\theta}_i{}^2 \\ \dot{\theta}_j{}^2 \end{bmatrix} + \begin{bmatrix} D_{iij} & D_{iji} \\ D_{jij} & D_{jji} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \dot{\theta}_2 \\ \dot{\theta}_2 \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} D_i \\ D_j \end{bmatrix} \tag{8.23}$$

The term D_{ii} is the effective inertia at point i and D_{jj} is the effective inertia at point j. The terms D_{ij} and D_{ji} are the coupling inertia terms. The terms in the second matrix on the right hand side represent the centripetal forces acting on the respective links. The terms in the third matrix represent the Coriolis forces acting on the robot. The remaining terms represent the effects of gravity on the links.

REFERENCES

[1] Niku, S.B., *Introduction to Robotics*, Pearson Education, 2011.

[2] Sciavicco, L. & Siciliano, B., *Modelling and Control of Robot Manipulators*, Springer, 2001.